



Hackney Ladish

708 South Elmira Avenue - Russellville, AR 72802 Phone: (800) 527-4500

Mill Test Report (According To: EN 10204:2004 TYPE 3.1)

Order Number: SOM881367

Sold To: STUART C IRBY COMPANY 815 IRBY DR PO BOX 1819 Jackson, MS, 39201, US

Customer PO: S014201970

Ship To: STUART C IRBY COMPANY 815 IRBY DR PO BOX 1819 Jackson, MS, 39201, US

Starting Material Heat #	Mfg. Heat Code	Item Description	SMLS / Welded	Material Spec Grade	Nace Information
18E103364	A19XHF1	12 X 10 XH TEE Y52 W/BARS	Seamless	MSS SP-75-2019 WPHY 52	NACE MR0103/ISO17945:2015 (R2023) NACE MR0175-2021/ISO15156-1:2020

Starting Material - Description	SMLS PIPE - ASTM A519-1518 MODIFIED
Starting Material Manufacturer	Yangzhou Chengde Steel Pipe Co., LTD, JIANGSU, CHINA, China
Forming Location	HACKNEY LADISH, INC., Russellville, Arkansas, United States
Heat Treatment Location	HACKNEY LADISH, INC., Russellville, Arkansas, United States
Heat Treatment Method	AUSTENITIZED AT 1650°F AND IMMEDIATELY IMMERSION QUENCHED IN WATER THEN TEMPERED AT 1250°F, Temper cycle held at temperature for 1 hour per inch of cross section (one hour minimum) and cooled in still air, BARS STRESS RELIEVED AT 1100°F

	Chemical Composition %															Carbon Equivalent				Hardness		Production Hardness #1		Production Hardness #2	
	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	V	Nb	Al	Ti	B	N	IIW	MSS*	PCM	CSA	Base	Weld	Base	Weld	Base	Weld
Ladle	.150	.250	1.200	.008	.002	.100	.100	.070	.030	.054	.003	.0290		.0004	.0039	.40	.38	.24	.38	204 HB					
Product	.140	.210	1.210	.013	.009	.100	.120	.080	.020	.050						.40	.37	.23	.37						

* Applicable Carbon Equivalent calculation in accordance with the above material specification edition specified.

Impact Test																	Tensile/Yield Strength Tests							
Specimen Size	Type	Ntch	Temp	Ft LB	Ft LB	Ft LB	Avg Ft LB	Mils Exp	Mils Exp	Mils Exp	Avg Exp	% Shear	% Shear	% Shear	Avg Shear	Loc	Base Material - T .252 IN (Diameter)				Welded -			
																	Yield	Tensile	%RA	%E	Yield	Tensile	%RA	%E
																	67 Ksi	88 Ksi		28				
10 X 10	TCVN	ANSVER	20 °F	160	130	130	140	64	64	72	66	75	70	100	81	Base								

Additional Notes MT Bar Welds / Certificate: MT967154-1 / Result: Satisfactory

We certify that the material herein described has been manufactured in accordance with the above standards, specifications, and purchase order and satisfies all the requirements of the editions specified. ISO 9001:2015 certified. The above figures are correct as contained in the records of the Company. This information has been electronically transmitted to our customer. Mark Ramey / Quality Analyst

Quality Analyst

Date: 03/31/2025